Question			Marking details	Marks available					
				AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
3	*(a)*		$I = \frac{12.0\ V}{450\ \Omega + 150\ \Omega}$ $[= 0.020\ \mathrm{A}]$ or equivalent **or** $\frac{V_{\mathrm{out}}}{V_{\mathrm{in}}} = \frac{150\ \Omega}{450\ \Omega + 150\ \Omega}$ or equivalent (1) Give mark if included in single equation (as below) $V_{\mathrm{out}} = 0.020\ A \times 150\ \Omega$ **or** $\frac{150\ \Omega}{450\ \Omega + 150\ \Omega} \times 12.0\ \mathrm{V}$ **and** = 3[.0 V] [as required] (1)		1	1	2	2	
	(b)		Any valid power calculation i.e. $\frac{3^2}{150}$ or $\frac{9^2}{450}$ or $\frac{12^2}{600}$ (1) **For 2nd mark:** - If used $\frac{9^2}{450}$ power in R_1 is 0.18 W which is < 0.5 W **or** which meets requirement - If used $\frac{12^2}{600}$ total power is 0.24 W and therefore power in R_1 is < 0.5 W (1) If $\frac{3^2}{150}$ don't award 2nd mark **ecf** on current and V_{out} from part (a)		1	1	2	2	
	(c)		$\frac{1}{R} = \frac{1}{150} + \frac{1}{600}$ or equivalent or by implication (1) $R = 120\ [\Omega]$ (the modified R_2) (1) $V_{\mathrm{out}} = 2.5\ [\mathrm{V}]$ **or** $2.53\ [\mathrm{V}]$ (1) So drop of > 0.4 V [therefore requirement not met] (give mark only if whole argument valid) (1) **ecf** on V_{out} from part (a)		1 1 1	1	4	2	
			Question 3 total	**0**	**5**	**3**	**8**	**6**	**0**